

User-Defined Interaction Using Everyday Objects for Augmented Reality First Person Action Games

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ABSTRACT

In this paper, we present an elicitation study to explore how people use everyday objects for augmented reality first person games. 24 participants were asked to select items from a range of everyday objects to use as controllers for three different classes of virtual object. Participants completed tasks using their selected items and rated the experience using the Augmented Reality Immersion (ARI) questionnaire. Results indicate no strong consensus linking any specific everyday object to any virtual object across our testing population. Based on these findings, we recommend developers provide the ability for users to choose the everyday objects they prefer.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality, Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies, Human-centered computing—Interaction design—Interaction design process and methods—Participatory design

1 INTRODUCTION

Creating interaction methods with virtual content in Augmented Reality (AR) and Virtual Reality (VR) which are both enjoyable and efficient remains an ongoing challenge. While research shows direct physical interaction with virtual content can lead to increased immersion as well as faster and more accurate performance [2, 4], the difficulty of implementing high quality physical interaction leads to many developers resorting to indirect speech or gesture interaction instead [1].

Tangible User Interfaces (TUI) are one type of interface that link physical objects to virtual objects for interaction. Historically, TUIs have used physical devices specifically designed for high quality motion tracking, however there is increased interest in exploring the use of everyday objects because this would avoid the need for (often expensive) dedicated hardware, and would take advantage of the many objects that the user is likely to already have at their disposal.

In this research, we explore the use of everyday objects for TUIs in the context of first person AR action games. This research will help to inform designers in this genre of games what types of objects their systems might need to recognise and track, and how interactions should be designed to support the preferred use of said objects.

2 METHODOLOGY

We conducted a user study based on the elicitation study methodology described by Wobbrock et al. [7]. Elicitation studies are a participatory design method which define a set of interactions based on proposals by participants. Participants are shown a set of “referents”, typically functions an application should perform, and asked

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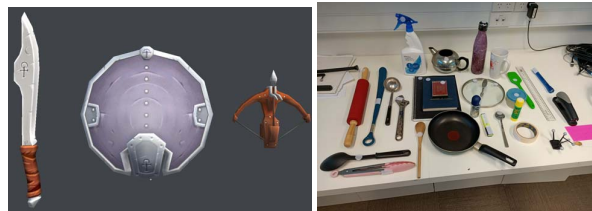


Figure 1: Left: The three virtual objects, from left to right: sword, shield and crossbow. Right: The range of everyday objects available for selection

to propose a method to elicit that referent. Similar proposals are then grouped, the most common proposal for each referent is defined as the “consensus set”, and an agreement rate can be defined to estimate how strongly the population agrees with this consensus set.

In this user study our referents are a set of virtual objects with behaviours which are widely understood and typical to first person action games: a sword, a shield and a crossbow, as shown in Figure 1 (Left). These three objects represent a range of common interaction types: the sword represents short range attacks, the shield represents defense, and the crossbow represents long ranged attacks. Each object had a task created for the user to complete: for the sword it was hitting four training dummies (see Figure 2 (Top)), for the shield it was blocking ten arrows from crossbows firing at the participant, and for the crossbow it was shooting at four training dummies using a voice command to fire (see Figure 2 (Bottom)). The experience was displayed using a Microsoft HoloLens 2 headset.

During the experiment, each participant was presented with a range of everyday objects, as shown in Figure 1 (Right), representing a subset of “The Full Directory Of Household Items”¹. The 500 most popular objects were considered, then all objects that might be impractical for the study were removed. Following this, objects with similar shape and function were grouped. Of the 100 entries remaining, as many objects were collected as practical, 26 total. For each virtual object, participants were asked to choose one everyday object to use as a controller for the virtual object. All of the everyday objects were tracked using an OptiTrack motion capture system.

To gain insight into the motivations behind their selections, each participant was asked to talk aloud during object selection and the everyday object that they selected was recorded. During the experiment, the object alignment and task was video recorded, and after each task the participant would rate their experience using the ARI questionnaire [3]. Once all tasks were completed, a short semi-structured interview was conducted.

3 RESULTS

For the user study, 24 participants were recruited, 10 identifying as female, 11 as male, 2 as non-binary and 1 as genderfluid. Half of all participants (12) were between 18-24 years old, 7 were 25-29 years old, 4 were 30-34 years old and 1 was 35-40 years old. All but one

¹https://adsy.me/household_items/

Table 1: Objects chosen for each virtual object. Note that in the interest of brevity, we have only shown objects with at least 3 votes.

Sword	Shield	Crossbow
Shoehorn 8	Pot lid 11	Tongs 3
Rolling pin 4	Fry pan 9	Spray bottle 3
Spatula 4		Tea pot 3
		Torch 3

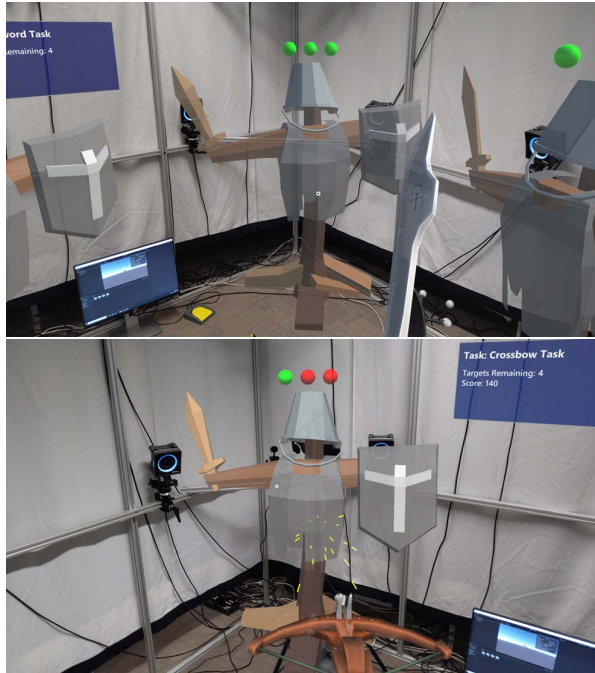


Figure 2: In game capture showing the sword task (Top) and the crossbow task (Bottom)

participant reported to having used a VR headset before, 13 reported to having used an AR head mounted display and 9 reported having used an AR head mounted display to play games before.

The objects selected by participants are shown in Table 1. Note that in the interest of brevity, we have only shown objects with at least 3 votes. The object with the most consensus for the sword was the shoe horn, with 8 selections and an agreement rate of 0.156, which Vatavu and Wobbrock define as medium agreement [6]. The pot lid was selected most for the shield with 11 participants and an agreement score of 0.330, representing high agreement. Finally, the crossbow had no consensus object with 4 objects being tied for 3 selections each and an agreement score of 0.050.

The results of the ARI questionnaire for the consensus and discarded sets were compared for the sword and shield object, the crossbow was excluded from further analysis due to no consensus being reached. One participant's response for the shield was also excluded due to tracking issues encountered during that task which may have affected their response. Statistical analysis indicated that there were no significant differences for any of the questionnaire factors between the consensus and discarded objects. This shows that the consensus objects don't necessarily provide a better experience, based on the factors of the ARI questionnaire, than the discarded objects.

4 DISCUSSION

Some aspects that might affect participant's immersion were identified through the post experiment interview. First, the limited field of view of the HoloLens was commented on by a number of participants, especially in the shield task which required them to look around. Future advancements in AR HMDs will likely improve FoV, while in the short term video see through VR headsets can provide a better FoV.

A second issue was the use of voice activation for firing the crossbow. Several participants mentioned how this felt unnatural, and we observed participants performing physical gestures alongside the voice command such as pulling a trigger or pulling back the crossbow string. Further work should be done to provide more intuitive ways of interacting rather than relying on voice controls.

The results of our study indicate that there is no strong consensus between the everyday objects and virtual objects chosen, and that consensus does not necessarily provide a better experience compared to the alternatives. We propose that it may be better to design systems that can recognise a range of everyday objects and allow the user to choose their own physical controllers. Alternatively utilising systems that aim to fit virtual objects to real world objects, like this system described by Hettiarachchi & Wigdor [5], could also provide a good alternative depending on the designer's intentions.

5 CONCLUSION

To explore the use of everyday objects in first person AR games, 24 participants took part in an elicitation study where they selected everyday objects to control virtual objects common in first person games. While two of the objects, the sword and shield, achieved reasonable consensus, there was no significant difference in the experience by participants who chose the consensus object over other objects. With a lack of strong consensus and evidence that the consensus provides a better experience, we recommend systems be designed to provide users the flexibility to choose their own objects.

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